AGRO PROCESSING /FOOD PROCESSING

Last Five Previous Year Question (2019-2023)

1. What is the major practice followed to preserve fish for long time?

- **a.** Salting and smoking
- **b.** Hulling
- **c.** Sorting
- **d.** All of the above

2. Hulling is an important processing practice followed in processing of which of the following crops?

- **a.** Pulses
- **b.** Millets
- **c.** Rice
- **d.** Wheat

3. __________ is the separation of the fruit and vegetable on the basis of quality parameters

- **a.** Sorting
- **b.** Grading
- **c.** Branding
- **d.** None of the above

4. High sugar preservation method is used to preserve which of the following produce

a. Vegetables
b. Cereals
c. Pulses
d. Fruits

5. Wet milling is the method of milling followed in which of the following crop

a. Rice

b. Pulses

c. Wheat

d. Millets

6. _____________ is the method of adopted to preserve the natural colour and flavour of meat products

a. Pan boiling

b. Frying

c. Curing

d. All of the above

Answers

1. A 2. C 3.B 4. D 5. B 6.C

AGRO PROCESSING /FOOD PROCESSING

Engineering inputs are vital for **modernization of agriculture, agro-processing and rural living**. It is needed for **development** and optimal utilization of natural resources, appropriate mechanism of unit operations of **agriculture for increasing production, productivity with reduced unit cost of production for greater profitability, economic competitiveness and sustainability.**

Agricultural produce and by-products are perishable in nature in varying degree and their perishability gets exploited on the market **floor compelling distress sales orchestrated** by **factors of demand and supply**, intervention of the faces of marketing in the absence of matching post-harvest technology (PHT) and agro-processing infrastructure.

Agricultural Engineering inputs are also needed to **assure remunerative prices** to the growers and a share in the value addition to the growers **through on-farm PHT and value addition to their produce and by-products** in order to strengthen their livelihood base landholdings are decreasing for their **socio-economic sustenance and assure minimum standards of living.**

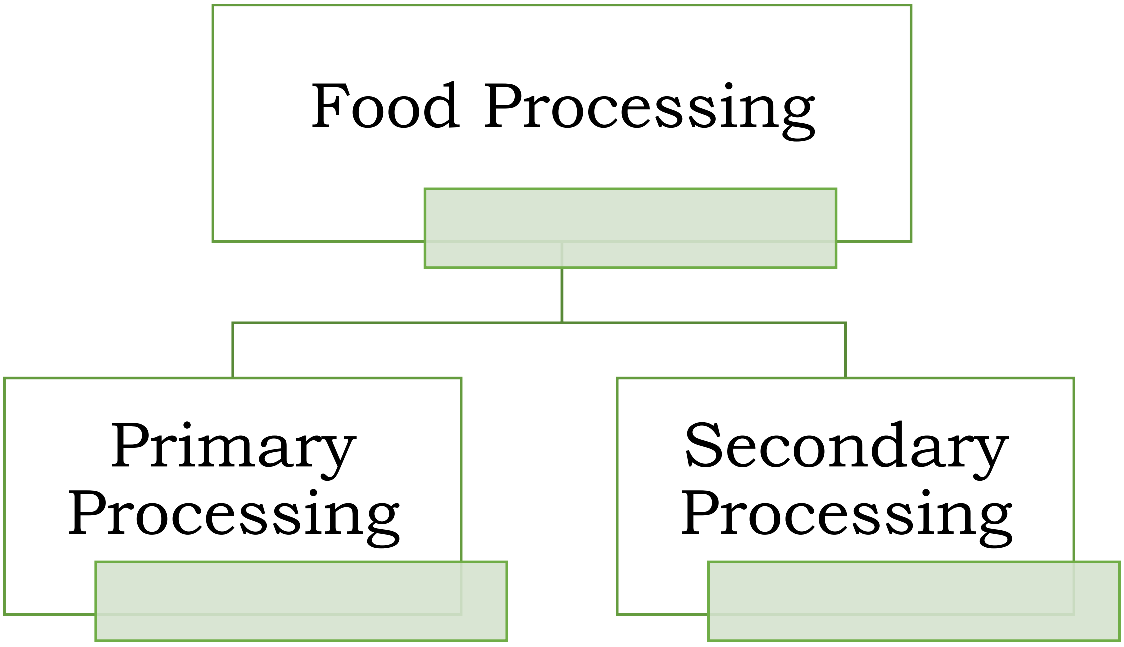
Food Processing

Food processing operations includes **many methods that are used to add value** to the **raw food materials** (including marine products, poultry and meat). **Raw food materials are transformed into edible products processing and value addition**. This provides **employments to rural people including women and prevents capital drain** from rural to urban areas and thereby helps in narrowing down the economic disparity between rural & urban population.

The operations involved in food processing are mainly classified into two groups, viz., primary processing and secondary processing.

Primary Processing –

Relates to **conversion of raw agricultural produce, milk**, meat, fish cereals and pulses etc into a commodity that is fit for human consumption. **It involves steps such as cleaning, grading, sorting, packing, etc.**



Secondary Processing -

Industries usually **deal with higher levels of processing** where new or modified food products are manufactured **ex ready to eat food, sauces, jam, jelly, snacks etc.**

Present status of Food Processing

Food Processing (FP) sector emerged as an important segment of the Indian economy in terms of its contribution to GDP, employment and exports.

During the last seven years ending 2021-22, FP sector has been growing at an **Average Annual Growth Rate (AAGR) of around 7.26%.**

Gross Value Added (GVA) in FP sector has also increased from **1.30 lakh crore in 2013-14 to 2.08 lakh crore in 2021-22.**

To meet the current demand of food materials, the industrial food processing sector has emerged. The food processing sector in the country is mainly handled by the unorganized sectors.

The **small-scale food processing sector** is a **major source of employment** and adds **value to crops by processing**. It is a major source of food in the human diet.

Importance of Food Processing

All the raw food materials are processed to **improve their palatability**, **nutritional value and shelf-life**.

Foods are processed for five major reasons:

- **Preservation f**or later consumption or sale to fetch better price.
- **Removal of inedible portions**.
- **Destruction or removal of harmful** substances.
- **Conversion to forms desired by the consumer**.
- **Subdivision into food ingredients**.

Scope of Food Processing

India is the world's second largest producer of food next to China, and has the potential of being the biggest with the food and agricultural sector. The **total food production in India is likely to double in the next ten years and there** is an opportunity for large investments in food and food processing technologies, skills and equipment, especially in areas of **Canning, Dairy and Food Processing, Packaging, Frozen Food/Refrigeration and Thermo-Processing. Fruits & Vegetables Processing**, Fisheries, Milk & Milk Products, Meat & Poultry, Packaged/Convenience Foods, Alcoholic Beverages & Soft Drinks and Grain processing are important sub-sectors of the food processing industry.

The consumer product groups like confectionery, chocolates and cocoa products, Soya-based products, mineral water, high protein foods, soft beverages, alcoholic and non-alcoholic fruit beverages, etc. along with the health food and health **food supplements is another rapidly rising segment of this industry** which is gaining vast popularity.

India produces **nearly 16% of the world's total food grain production**. It is one of the largest producers of agricultural produce. With a population expected to reach to about **590 million people by 2030 in urban India, India has a huge potential domestic demand for processed foods other than the demand from the exports**.

There are many **socio-economic factors** that are driving the demand side of the Indian Food Processing Industry.

The **changing consumption patterns**, both in tier 1 and tier 2 cities, **rising income levels** among the middle-class and **changing lifestyles**, are some of the factors providing the demand side push for the Food Processing Industry. Moreover, the central government has given a priority **status to all agro-processing** businesses.

Key Constraints for Growth

There are still some significant constraints which**, if not addressed properly, can impede the growth prospects of the Food Processing** Industry in India.

- **One of the biggest constraints is that this industry is capital intensive**. It creates a strong entry barrier and allows lesser number of players to enter the market
- **Lesser competition** and lesser competition mean reduced efforts to improve the quality standards.
- **Poor infrastructure** for storing raw food materials**. Storage system** like the warehouses and the cold storages, **lag in storage standards.**
- **The pests infest the grains** sometimes due to lack of monitoring, proper use of pesticides and proper ventilation
- **The power outages** result in sub-optimal function of the cold-storages and the quality of food material in the cold storages becomes questionable.
- **The second important aspect is having poor quality standards** and **control methods for implementing the quality standards** for processing and packaging the processed foods. **For example**, vegetables may not be washed properly and processed into either 'ready to eat food' or packaged as 'cut and ready to cook' vegetables.
- **Continuity of quality power, good quality of water for processing**, instruments for rapid and reliable analysis, versatile instruments/equipment's for multi commodity, cultivars **are not suitable for specific processing**, etc. are other limitations for food processing industry.

Methods and Principles of Food Processing

Food processing therefore **refers to the application of techniques to foods in a systematic manner for preventing losses through preservation, processing, packaging, storage and distribution**, ultimately to ensure greater availability of a wide variety of foods which would help to improve the food intake and nutritional standards during the periods of low availability.

The main objective of fruits and vegetables processing is to supply wholesome safe, nutritious and acceptable food to consumers throughout the year.

Prerequisites Of Preservation Are Cleaning, Grading and Sorting as Per Maturity

1. **Cleaning**- It is a unit operation in which **contaminating materials are removed from the food material and separated to leave the surface of the food** in a suitable condition for further processing. In addition, the early removal of small quantities of food contaminated by micro-organisms **prevents the subsequent loss of the remaining bulk by microbial growth during storage** or delays before processing.

Types Of Cleaning

Wet cleaning- Washing Agri **produce with water for removing soil from root crops** or dust and **pesticide residues from soft fruits or vegetables**. It is also **dustless** and causes less damage to foods than dry methods.

Figure 1 – Wet Cleaning

Dry cleaning: Dry cleaning procedures are used **for products that are smaller**, have **greater mechanical strength and possess lower moisture content** (for example grains and nuts). After cleaning, the surfaces are dry, to aid preservation or further drying.

The main groups of equipment used for dry cleaning are- air classifiers magnetic separators, separators based on screening of foods

Figure 2 – Dry Cleaning

2. **Sorting**

Sorting is the separation of foods into categories on the basis of a measurable physical property. Like cleaning, sorting should be employed as early as possible to **ensure a uniform product for subsequent processing**.

The four main physical properties used to sort foods are size, shape, weight and colour.

a. **Shape and size sorting**: The **particle size distribution** of a material is expressed as either the **mass fraction of material** that is retained on each sieve or **cumulative percentage of material retained**. Size sorting is the separation of solids into two or more fractions on the basis of differences in size.

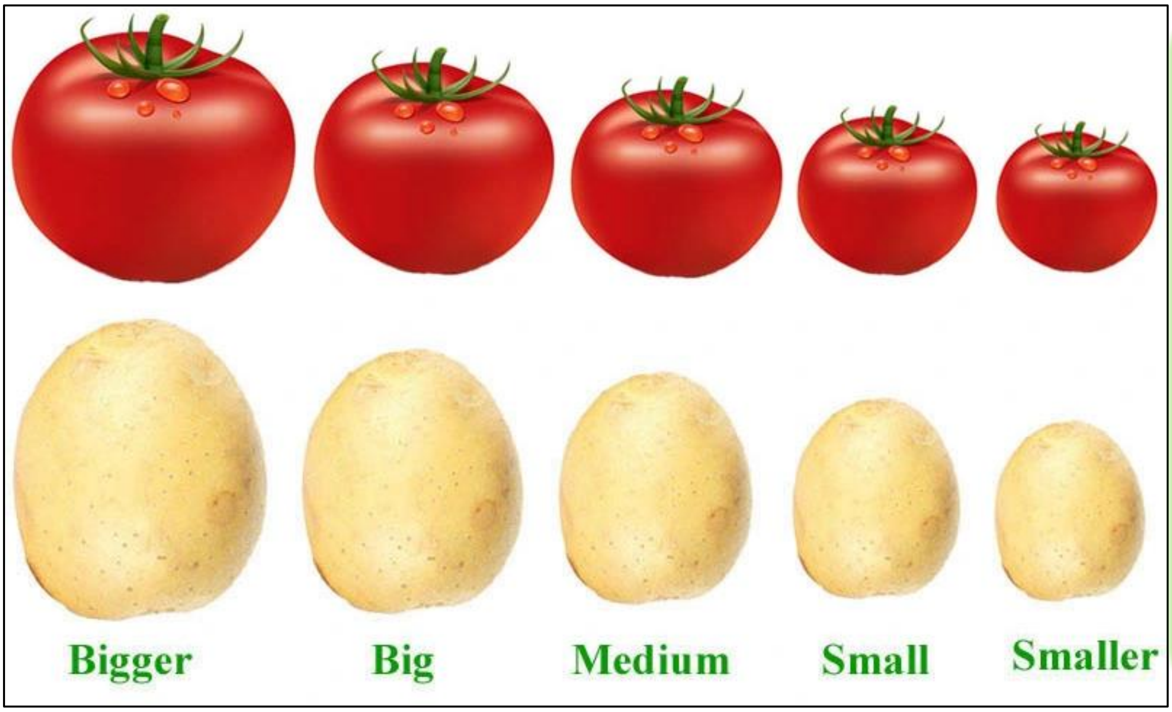


Figure 3- Sorting Of Tomato and Potato on The Basis of Size

b. **Colour sorting**: Small particulate foods may be **automatically sorted at high rates using microprocessor-controlled colour sorting equipment**. Food produce is made to pass through a photo detector. **Photo detectors measure the reflected colour of each piece and compare it with preset standards**, **and defective foods are separated by a short blast of compressed air.**

c. **Weight Sorting**: Weight sorting is **more accurate than other methods** and is therefore used for more valuable foods. **Aspiration and flotation sorting use differences in density to sort food** and are similar in principle and operation to aspiration and flotation cleaning.

3. **<u>Grading:</u>** This term is often used interchangeably with sorting but strictly **means 'the assessment of overall quality of a food using a number of attributes**. Grading is carried out by machines or operators who are trained to simultaneously assess a number of variables.

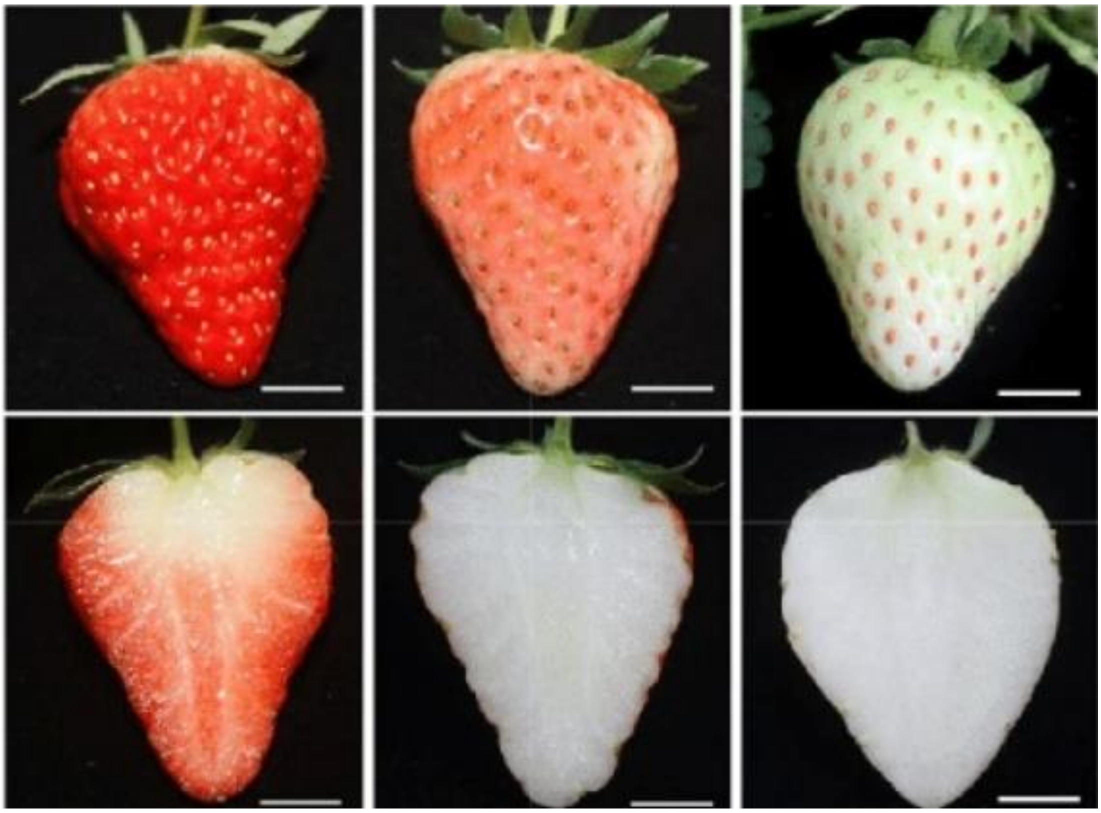


Figure 4 – Sorting Of Strawberry on The Basis Of Colour

Figure 5 – Grading Of Apple on The Basis of Quality

The methods and principles can further classify based on food preservations. These are;

1. **Control of Water Activity**

Water activity of the food can be controlling the different methods which are given below

a. **Evaporation**- A **raw material or a potential foodstuff contains more water than** is **required** in the final product. When the foodstuff is a liquid, the easiest method of **removing the water, in general, is to apply heat to evaporate it**. Evaporation is thus a process that is often used by the food technologist. **Changes that may occur in the foodstuff during the course of the evaporation process.**

b. **Dehydration:** It is defined as the **application of heat under controlled conditions to remove the majority of the water normally present in a food by evaporation**. The main purpose of dehydration is to extend the shelf life of foods by reduction in water activity.

Figure 5 – Preparation Of Dried Apricot

c. **Osmo-Dehydration**: In osmo-dehydration, the prepared **fresh material is soaked in a thick liquid sugar solution and or a strong salt solution and then the material is dried**. During osmotic treatment, the material loses some of its moisture.

- The syrup or salt solution has a protective effect on colour, flavour and texture.
- **This protective effect remains throughout the drying process and makes it possible to produce dried products of high quality.**

d. **Drying:** Drying is the process **to remove the moisture content from the food material** and thereby reduce the water activity and extend the shelf life. Several types of

dryers and drying methods are commercially used. This is depending upon the form of material and its properties, desired physical form and characteristics of dried products, operating cost, etc. **Sun drying, solar drying, atmospheric drying (batch type) like, kiln, tower, cabinet and (continuous type) like tunnel, belt, fluidized bed, puff, foam mat, spray, drum and microwave, etc**.

Figure 6 – Sun Drying

2. **Cold Treatment**

Cold treatment is the **treatment like, fermentation, irradiation, di-electric, ohmic, infrared heating, freezing, super cooling, refrigeration**, etc. given to the food materials for increasing the shelf life.

Figure 8 – Fermented Drink (Kanji)

A. Fermentation:

During food fermentations, **the controlled action of selected micro-organisms is used to alter the texture of foods, preserve foods** by production of acids or alcohol, or to produce subtle Flavors and aromas which increase the quality and value of raw materials.

B. Irradiation:

Ionizing radiation takes **the form of ϒ-rays from isotopes** or, commercially to a lesser extent, **from X-rays and electrons**. There is **little or no heating of the food** and therefore negligible change to sensory characteristics.

Applications of Irradiation

- Sterilization
- Reduction of pathogens
- Prolonging shelf life
- Control of ripening

- Disinfestations
- Inhibition of sprouting

C. Dielectric, Ohmic and Infrared Heating:

Dielectric energy and infrared energy are two forms of electromagnetic energy. They are both transmitted as waves **which penetrate food and are then absorbed** and converted to heat. In contrast, **ohmic heating uses the electrical resistance of foods to directly convert electricity to heat**. Dielectric and ohmic heating are direct methods in which heat is generated within the product, whereas infrared heating is an indirect method that relies on heat that is generated externally being applied to the surface of the food mostly by radiation, but also by convection and to a lesser extent conduction.

D. Freezing:

Freezing is the **reduction in temperature generally by super cooling** followed by crystallization of water, nucleation and finely crystal growth.

Methods of quick freezing

1. **Freezing by indirect contact with refrigerant**: Food may be **frozen by being placed in a contact with a metal surface which is cooled** by a refrigerant or packaged or packed in a can and **cooled by immersion in a refrigerant.** Also, food packaged in paper boxes may be frozen by contact with refrigerated metal plate which may be moving or stationary.
2. **Air Blast freezing**: To **obtain very cold air, a blast of air is directed** through refrigerating coil. For greater effect, the cold air blast is **confined in an insulated tunnel**. The material to be frozen may be placed on a moving belt within variable of moved counter current and the air blast.
3. **Freezing by direct immersion (FBDI): FBDI in low temperature drying** was the beginning of quick freezing. **Since liquid are good heat conductors**, a product can be frozen rapidly by direct immersion in low temperature liquid for example brine and sugar solutions.
4. **Super Cooling**: Occurs **when temperature of water is lowered below the freezing point and crystallization does not occur**. The super cooling provides the means of determining the in-depth effect of a reduction in temperature relative to the initial freezing point.

5. **Refrigeration:** This is the process **by which heat is removed from a confined place** and material for the purpose of maintaining a lower temperature. The standard unit of generating heat capacity is 1 tone of refrigeration. This is derived on the basis of removal of **latent heat of fusion of 1 tone of water at 32° F or 0°C to produce 1 tons of ice.**

3. Control of Microbial Activity

Food materials can be stored for **longer period by controlling the microbial activities by pasteurization and sterilization** methods.

a. Pasteurization:

Pasteurization is a process of heat treatment used to inactivate enzymes and to kill relatively heat sensitive pathogenic microorganisms that cause spoilage. It is a mild heat treatment, usually **performed below 100°C**. Pasteurization kills only harmful microorganisms.

In practice, therefore, m**ost of the canned foods produced** locally in developing countries such as canned peas tomatoes, canned pineapple slices etc. **are heated within the package**.

There are two categories of pasteurization process:

a) **Low temperature long time (LTLT):** 62.7°C for 30 minutes

b) **High temperature short time (HTST)**: 71.7°C for 15 seconds

b. Heat sterilization:

Sterilization: In this process, **food is heated at a sufficiently high temperature (121°C) and for long time (10-15 minutes) to destroy microbial and enzyme activity.** As a result, sterilized foods have a shelf life of more than six months. **Higher temperature for a short time (140°C/3-4 seconds) is possible if the product is sterilized before it is filled into pre-sterilized containers in a sterile atmosphere**. This forms the basis of Ultra High Temperature (UHT) processing (also termed aseptic processing). **It is used to sterilize a wide range of liquid foods (fruit juices and concentrates, wine, etc.) and foods that contain small discrete particles (tomato products, fruit and vegetable soups).**

c. Blanching:

Figure 9- Blanching

Blanching is a **heat treatment given to a fruit or vegetable either in boiling water or microwave above 100°C**. Blanching is **used to destroy enzyme activity** in fruits and vegetables, prior to processing. **Blanching helps in several ways as it inactivates enzymes**, which prevents **undesirable changes in sensory characteristics** and nutritional properties that take place during storage, reduces the number of contaminating microorganisms on the surface of foods, leads to softening of vegetable tissues thus facilitating can filling and helps in removal of air from intercellular spaces.

4. Chemical Preservation

Use of chemical additives such as sugars, salt, acids, spices etc. to preserve the food

High sugar preservation

Figure 10- High sugar preservation of orange

- In the food preservation **with sugar, the water activity cannot be reduced below 0.70**.
- This value is sufficient for **bacteria, yeasts and molds inhibition but does not prevent osmophilic yeasts and xerophillic molds attack**.
- For this reason, various means are used to avoid mould development such as **finished product pasteurization (jams, jellies, etc.)** and use of chemical preservatives.

Use of salt/acid/spices (Pickling):

Pickle is an edible product preserved and flavoured in a solution of common salt and/or vinegar.

- The **preservation of fruits and vegetables in common salt and/or in vinegar is called pickling.**
- **Spices and edible oils may be added** to the product.
- **Raw mango, lime, turnip, cabbage, cauliflower etc**., are preserved in the form of pickles, which have become popular in several countries.
- Apart from having nutritional and therapeutic value, they have appetizing appeal

Figure 11- Pickling Of Vegetables In Vinegar

Use of chemical additives:

The Food and Drug Administration (FDA) has defined food additive as a substance or a mixture of substances, other than the basic foodstuff, which is present as a result of any aspect of production, processing, storage or packaging.

- It comprises of preservatives, antioxidants and many others.
- **Chemical food preservatives are added in very small quantities (up to 0.2 per cent)** and they do not alter the organoleptic and physico-chemical properties of the foods.
- **Ex- sodium benzoate, benzoic acid, potassium nitrate**, sodium nitrate generally potassium nitrate and sodium nitrate are used for meat products not for fruit and vegetable products.

C. Combined Method of Preservation (Hurdle Technology)

Judicious combination of more than one method for synergistic preservation is called hurdle technology.

- **Hurdle in food is defined as the substance or the processing step or various preservation factors, inhibiting** the growth of various microorganisms resulting in the death of microorganisms.
- **But loss of produce quality is reported in various fruit and vegetables.**

VALUE ADDITION AND PROCESSING OF HORTICULTURAL CROPS

Processing Of Fruits and Vegetables

1. Jam, Jelly, Marmalade and Preserve:

Preparation of jam, jelly and marmalade is based on concentrating **fruits to nearly 70 per cent solids (TSS) by addition of sugar and heat treatment**. The high osmotic pressure of sugar creates unfavourable conditions for the growth and reproduction of most species of microorganisms i.e. yeasts, moulds and bacteria, responsible for the spoilage of food. **At this concentration of solids, the water activity is reduced (aw of 0.60-0.75), which ultimately decreases the chances for microbial spoilage.**

A. Jam

Jam is prepared by boiling the fruit pulp with sufficient quantity of sugar to a reasonably thick consistency, firm enough to hold fruit tissues in position. It should **contain not less than 68.5 per cent soluble solids** as determined by a **refractometer**.

Figure 11- Mango Jam

- Jam may be **made from a single fruit** (apple, strawberry, banana, pineapple etc.) or from a combination of two or more fruits.
- The **preparation of jam requires several unit operations** viz., selection of fruit, preparation of fruit, addition of sugar, addition of acid, mixing, cooking, filling, closing, cooling and storage.

B. Jelly

It is a **semi-solid product prepared by boiling a clear, strained solution of pectin containing fruit extract with sufficient quantity of sugar and measured quantity of acid**.

Figure 12 – Fruit Jelly

- A perfect jelly should be **firm enough to retain a sharp edge** but should be tender enough to resist the applied pressure.
- Different fruits like **guava, plum, papaya, gooseberry etc. are used for jelly preparation. Low pectin fruits such as apricot, pineapple, raspberry etc. can be used only after adding small amount of pectin powder.**
- The essential substances for **manufacture of jelly are pectin, water, acid and sugar**. **Formation of jelly takes place when the concentrations of water sugar-acid-pectin mixture attain a certain minimum value**

C. Marmalade

It is a fruit jelly **in which slices of the citrus fruit or its peels** are suspended.

Marmalades are generally **made from citrus fruits like oranges and lemons** in which shredded peels are suspended.

Figure 13 – Marmalade

D. Preserves (Murabbas)

These are prepared from **whole fruits and vegetables or their segments** by addition of sugar followed by evaporation to a point where microbial spoilage can't occur. **The final soluble solids concentration reaches to about 70 per cent**. The finished product can be stored without hermetic sealing and refrigeration.

Figure 13 – Amla Murabba

2.Chutneys and Sauces

Chutney is a mixture of fruit or vegetable with spices, salt and/or sugar, vinegar etc. A good chutney is **smooth, palatable and appetizing**, and has the true single flavour of the fruit or the vegetable used for its preparation.

Figure 14 – Chutneys

- **Vinegar, salt, sugar and spices are the common preservatives**, used for the preservation of these products.
- The chemical preservatives, such as s**odium benzoate and potassium metabisulphite are used for long-term storage,** which helps in retarding the growth of microorganisms without interfering with other physio-chemical and sensory characteristics of the product

3.Fruit Juices/Beverages

Fruit juices are preserved in different forms such as pure juices and beverages.

Fruit beverages can be classified into two groups –

- Fermented Drinks
- Non-Fermented Drinks

1. **Unfermented Beverages**:

Fruit juices that **do not undergo alcoholic fermentation are termed as unfermented beverages**.

- They include **natural and sweetened juices, ready-to-serve beverage**, nectar, cordial, squash, crush, syrup, fruit juice concentrates and fruit juice powder.

- These beverages can be distinguished on the basis of the **differences in total soluble solids (TSS) content and minimum juice percentage**

Figure 15- Mango Squash

2. **Fermented Beverages**:

Fruit juices, which have **undergone alcoholic fermentation by yeast** and lactic acid fermentation by bacteria. They include **wine, champagne, port, sherry, cider**.

Preservation Of Fruit and Vegetables

A. **Pasteurization:** Preservation **by heat is the most common method** of fruit juice preservation. It may be done in three ways:

B. Carbonation: It is the process of **incorporating carbon dioxide in a beverage** to impart a characteristic taste. Apart from the distinctive taste, carbon dioxide also inhibits the growth of certain undesirable microorganisms.

C. Chemical Preservation: For preserving juices chemically, the addition of **700 ppm potassium metabisulphite or 720 ppm of sodium benzoate** (for coloured products) is employed. Chemically preserved

Figure 16 – Coke, Carbonated Drink

juices are bottled leaving **a head space of 1.5 to 2.5 cm followed by crown corking**/sealing

D. Fermented products: Fermentation of fruits and vegetables can be classified into three types:

1. **Alcoholic fermentation**

2. **Lactic fermentation** – which involves **fermentation of carbohydrates into lactic acid** to prepare fermented pickles such as sauerkraut, gherkins, fermented olives etc.

3. **Acetic fermentation** – It involves **alcoholic fermentation followed by acetic acid fermentation for the manufacture of vinegar, which is used as a condiment**. Vinegar contains **about 4 per cent of acetic acid** in water and can be prepared from a number of fruits such as grapes, apple, oranges etc

E. Pickles:

The process of **preservation of food in common salt or in vinegar is called pickling**. Pickles may be **sour, sweet or mixed** and can be prepared easily from different fruits and vegetables at home.

Figure 17 – Lemon Pickle

- They can be **grouped as unfermented pickles and fermented pickles**.
- **Fermented pickles undergo lactic acid fermentation** on the other hand, in **unfermented pickles, the raw material is preserved by use of various spices and oil.**
- Most popular unfermented pickles are **mango, lime and mixed pickles**.

E. Dried products: Fruits or vegetables may **be dried mechanically or under the sun for increasing their shelf life and for further use.**

- **Grapes are dried and converted into raisins**.
- Dehydrated powders of **various citrus fruits are available for reconstitution into a refreshing beverage**.

- **Onions and ginger are sold in dehydrated form** for use in various curried food preparations

PROCESSING OF FLORICULTURAL CROPS

1. Dry flowers and pot pourri:

Dry flowers are becoming more **popular due to their longer indoor life and non-perishability.** The two easiest and least expensive methods to dry flowers are **sand drying and air-drying**.

- **Pot Pourri** is a mixture of **dried, sweet scented plant parts including flowers, leaves, seeds, stems and roots. These are rich in aromatic oils**, which are not confined to the flower only. These are used **in naturo-therapy** for common ailments (aromatherapy).
- **Fixatives such as salt, gum benzoin etc.,** are added to make the scent to last longer

Figure 18- Sun Drying of Flower

2. Essential oils, flavours and fragrances:

Floral extracts like essential oils, alkaloids, pigments, dyes etc., have tremendous demand in both domestic and international markets. In order to produce the highest quality extracts, **highly sophisticated extraction methods such as those based on high pressure extraction and super critical fluid extraction are used.**

- Such methods produce **very high purity flavour and spice extracts**, fragrant chemicals as well as pharmaceutical substances.
- They have the **density of a liquid but diffuse as a gas and therefore, function like a solvent**. This enables the extraction of sensitive raw materials at gentle temperatures.

Figure 19 – Essential Oils

- The resulting extracts are **further purified by fractionation and separation** procedures.

3. Pharmaceutical and nutraceutical products:

Plants produce pharmacologically valuable compounds, which **are used in medicine and as dietary supplements.** Such compounds include **pigments, oils and alkaloids**.

4. Pigments and natural dyes:

The **anthocyanins, flavanols, carotenoids and xanthophylls are common plant pigments t**hat are responsible for a variety of hues we normally observe. These valuable **pigments can be isolated** and used for varied applications including pharmaceuticals.

Figure 20- Natural Dyes

Gulkand, rose water, vanilla products etc. Gulkand is prepared by mixing rose petals and sugar in the ratio of 1:2 followed by mashing and drying the mixture in sun. **Rose water is prepared by boiling the rose flowers** in water and condensing the steam. It is used as **sherbet, eye lotion and eye drops**.

5. Insecticidal and nematocidal compounds:

Natural plant products (**secondary metabolites) are insecticides and nematicides**. They act as fly and **mosquito repellents, kill insects** and may be toxic to bees, aphids, caterpillars etc.

PROCESSING OF FARM CROPS -CEREALS AND PULSES

Cereal grains are hygroscopic and would gain or lose moisture initially until they are in equilibrium with air. **Food grains include cereals like rice, wheat, maize, sorghum**

and millets; pulses like pigeon pea, chick pea, black gram and oilseeds like groundnut, mustard, coconut, sesame.

Storage is done to meet the food, feed and seed requirements of the people between two harvests and during natural **calamities like draught, famine, war** etc.

Milling of food grains and oil seeds is done to convert them into suitable products.

Processing Of Cereals

1. Rice (Oryza sativa L.)-

A paddy grains consists of several layers of components. **The major components of paddy grain are the husk, bran layers, starchy endosperm and the embryo or the germ.**

- **Rice husk or hulls** are the coating for the seeds. The hull is indigestible to humans.
- **Bran is the hard outer layer of grain and consists of combined aleurone, nucellus, seed coat and pericarp**. Along with germ, it is an integral part of whole grain, and is often produced as by-product of milling
- The **embryo or the germ is rich in protein** and is often removed along with bran during the milling and polishing process.
- The **starchy endosperm is the popular white rice.** A paddy grain approximately contains **20 % by weight of husk, 5 to 7 % by weight of bran layers and about 3 %** by weight of embryo.

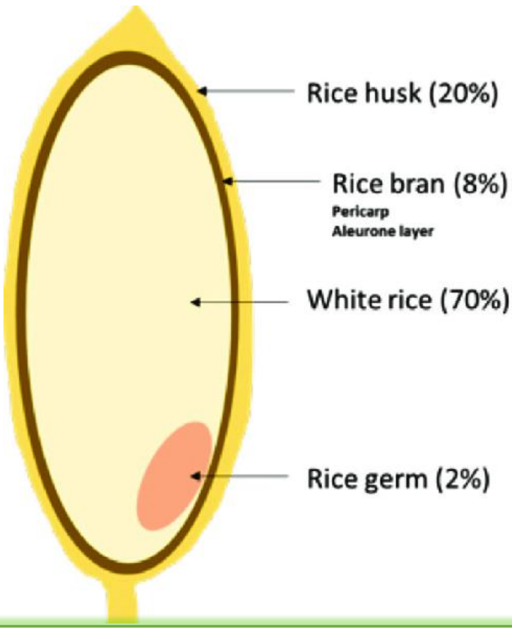


Figure 21- Structure Of Rice

Processing

Rice milling is the process **wherein the rough rice grain is transformed into a form suitable for human consumption,** therefore, has to be done with utmost care to prevent breakage of the kernel and improve the recovery.

Methods Of Milling – Hand Pounding, Milling Etc

Parboiling

Parboiling is an **optional and pre-milling process given to the paddy**. It is hydrothermal process in **which paddy soaked in water followed by gelatinization by steaming**.

- The step process involves **soaking, steaming and drying.**
- There are different methods in parboiling su**ch as household method, single steam method, double steam method, hot water soaking method and pressu**re parboiling method etc.

Figure 22- Parboiled Rice and White Rice

- **Among these methods single steaming, double steaming** and hot soaking methods are adopted in commercial scales.

Hulling

When only the **husk layer is removed from paddy grain the resulting product is called brown rice and the process is called as hulling.**

- Brown rice is milled further to remove bran layers for creating a **visually more appealing white rice than the brown rice.**

Milling

The **dried paddy is milled to produce rice**.

In this method of **milling the husk is removed by rubber roll sheller** and the bran and germ portions of rice are separated during polishing in the cone polishers.

- During milling**, the husk and bran layer are removed**. After removal of husk, the polishing is done to more than 5 % to obtain the write rice.
- Due to soft nature of kernels, the **raw milling results in higher percentages of broken**s with reduced out turn compared with parboiled rice.

- The milling **yield slightly varies and depends on the quality of raw paddy** being milled.

Figure 23 – Processing Of Rice

2. Processing Of Wheat

In the traditional method **of harvesting ripe wheat is cut and bound in to sheaves**, which **are stood in stacks in the field**. The action of **wind and sun reduces moisture of the wheat from between 16 percent in a few day**s. When the moisture is sufficiently reduced sheaves are collected, stacked and threshed. **The threshed grain with moisture of 14 % or less can be stored safely.**

Most important event, in **processing is milling of wheat, the term milling in reference** to the process involving food grains is a trade name used **for reduction of grains into flour or meal or any other consumable form for human beings or animals**.

In India, a large population **of wheat is used as atta and maida**.

The scope of milling covers a wide range **of process depending upon the grain and products.**

<u>The composition of bran germ and endosperm in wheat is given below</u>

- Bran – 12%
- Germ – 3%
- Endosperm – 85%

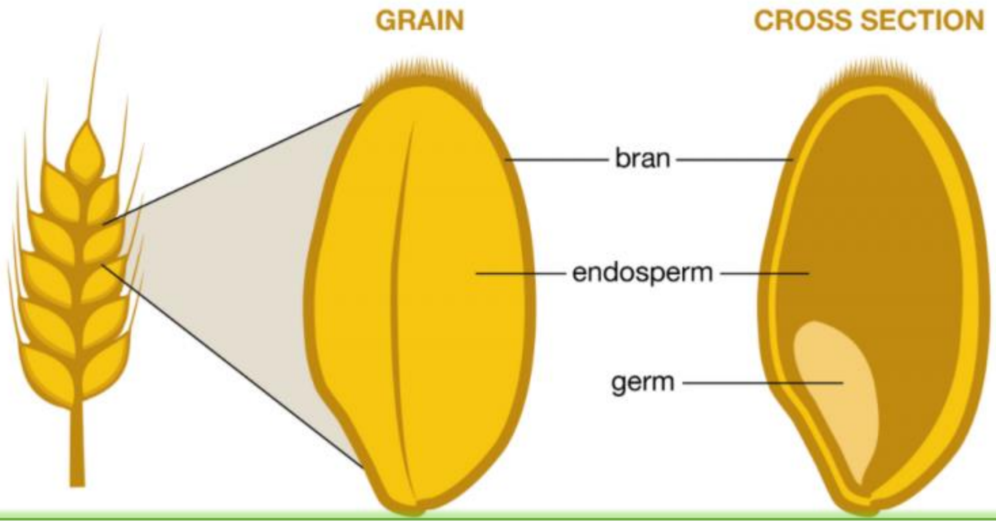


Figure 24 – Structure Of Wheat Grain

<u>Purpose of flour milling</u>

The purpose of flour milling is to first **separate the endosperm from bran and germ in large chunks as possible and then reduce the size** of the endosperm chunks flour sized particles through a series of milling steps. **Flour contaminated with wheat bran is brownish rather than white is not desirable for food purposes.**

Figure 25 – Milling Products Of Wheat

<u>Tempering or Conditioning</u>

Tempering refers to the addition of water to wheat grains before milling. The bran becomes tough and rubbery while the endosperm becomes less vitreous. **This improves milling efficiency.**

Conditioning, use of **heat for quick diffusion of water into endosperm** as well as the bran is the purpose.

Wheat grinding- The grinding **of wheat is done between pairs of rolls**. These rolls, since they are **moving opposite direction moving at different rate of speed one f**rom the other and set with an appreciable gap between Rather, the reduction of wheat size and turn it in to flour.

3. Processing Of Corn

Corn (Zea mays)

Corn is processed in to **different food and feed ingredients, beverages and industrial products.**

- It is usually harvested when its moisture content is in the **range of 18 to 24 %.**
- The maize kernels must be dried to safe moisture **levels of about 12 %** to avoid losses.

Milling Of Corn

Dry milling and wet milling are the most common methods are used for corn milling.

- **End products of dry milling process like germ** used for oil extraction while husk and de**-oiled germ used for cattle feed**, **grits for breakfast**
- **End products of starch, germ** and feed are the **products of wet milling**.

Dry Milling Method Of Corn

Dry milling method is divided in to two methods;

a. **Non-de-germing method (traditional**)- In traditional method, **the whole kernel is ground in to meal by stone grinder without removing germ** having high fiber and high protein. Then, the hull and germ are removed by sifting from the meal.
b. **De-germing method (modern)**. In de-germing method**, the kernel are moistened with a less amount of water and then tempered for equilibrium.** Then, the moistened **kernel are allowed to dry and then the hull, germ and tip cap are removed from the kernel to get corn grits**. The germ is used **for oil extraction** and de-oiled germ and hull are used for cattle feed purpose **which is known as hominy**

feed. The yield of endosperm products and **hominy feed are about 70 per cent and 30 per cent respectively.**

Rolling and Grading

The germ, husk and endosperm fragments are then separated by means of sifters, aspirators, specific gravity table separators or purifiers.

- **Sifting is** an important operation and is variously referred to as **scalping, grading, classifying, or bolting depending upon the means used and purpose.**
- **Sifting i**s actually a size separation ration on sieves.
- **Scalping is the coarse separation made** on the product leaving a roller mill or degermer. **Grading or classifying is the separation of a single stock** (usually endosperm particles into two or more groups according to particle size.
- **Bolting is removal** of hull fragments from a corn meal or flour.

Figure 26 – Milling Products Of Maize

Wet Milling method of corn

The **raw corn for wet milling should contain 15-16 per cent moisture** and it should be physically sound.

- Insect and pest **infested, cracked and heat damaged corns are unsuitable for wet milling**. The heat damaged corn affects the quality of oil extracted from its germ.

- Sufficient amount of moisture is added to the corn **during steeping in the wet milling process in order to prepare the corn for subsequent degerming, grinding and separation operations.**

4. Milling of Pulses

It is generally practiced **in Madhya Pradesh and Uttar Pradesh**. In this, the pulses are subjected to p**itting in a roller and then oil treatment by applying 0.5-2.0 per cent linseed oil or any edible oil.**

- Then the **pulses are spread in the drying yard** for **sun drying for 2 – 4 days**.
- The pulses are tempered **by heaping and covering during the nights** in between these days.
- After sun drying, **again pulses are moistened uniformly with about 5 % water** and kept as such on heaps overnight for moisture equilibrium.

Figure 27- Different Pulses Before and After Milling

- Then, these pulses are allowed to pass from the **roller for splitting and dehusking**. About, **50 % of the pulses are dehusked and split in first operation**.
- After this, **the husk are removed by aspiration and split dhal are separated** from the **mixture of husked and un- husked whole pulses**.
- The mixture is once again moistened **and dried in the sun and then dehusked and split**.

- This process of **alternate wetting and drying is repeated until almost all the remaining pulses are converted in to split dhal.**
- **The average yield of dhal ranges from 68-75 %.**

Processing Of Oil Seeds

Oilseed and nut should be properly dried before storage, and cleaned to remove sand, dust, leaves and other contaminants. All raw materials should be sorted to remove stones and moudly nuts. Some moulds, especially in the case of groundnuts, can cause aflatoxin poisoning.

Some raw **materials (for example groundnuts, sunflower seeds) need dehusking** (or decorticating). **Decortication is important to give high yields of oil** and reduce the bulk of material to processed.

Oil Extraction methods

a) Mechanical expression

During the process **of mechanical expression, the oil seeds are compressed in various types of compression devices/equipment**. Expression is the process of mechanically pressing liquid out of liquid containing solids. **Screw press, roll presses, collapsible plate are some examples of wide range of equipment used for expression of liquid.**

b) Oil Extraction

Extraction is a **process of separating a liquid from a solid system with the use of a solvent.** Extraction is also a **process of diffusion with the help of low boiling point solvent**.

- This process gives a **higher recovery of oil and a drier cake than expression**.
- **Solvent extraction is capable of removing nearly all of the available oil from oilseed** meal.
- This extraction process **provides meal of better preservation qualities** and with higher protein qualities.

c) Process of Oil Refining

In many local markets **further refining is not required as the complexes of unrefined oils are preferred. International markets tend to prefer lighter less intense** oils for

cooking which means further processing of the oil. There is serious of refining processes that can be carried out after the oil has been filtered.

d. De-odorising

Volatile **compounds that produce bad odours can eliminated through the process of sparging,** i.e. bubbling steam through the oil, under a vacuum.

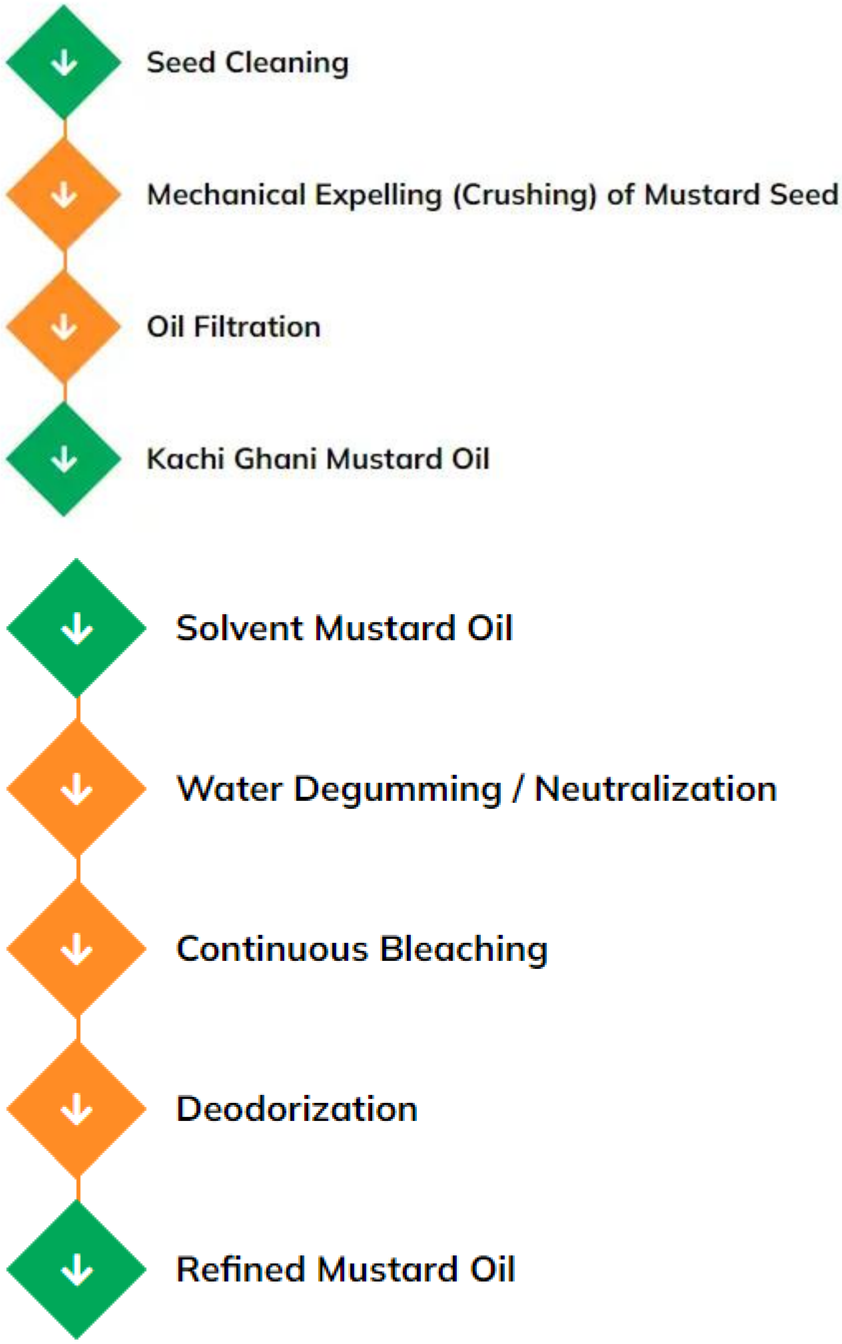


Figure 28 – Process Of Extraction Of Oil From Mustard Seeds

e. Wintering

Allowing the oil to stand for a time at low temperatures so that **glycerides, which naturally occur in the oil, with higher melting points solidify** and **can then be removed** from the oil by filtering. Over time glycerides can degrade releasing fatty acids into the oil increasing the acidity levels and reducing the quality.

f. Neutralisation

Fatty acids **can be neutralized by adding a sodium hydroxide solution**, also known as caustic soda, or by stripping, which is a similar process to de-odorising.

g. Bleaching

Some oils have a very dark colour to them that is unpopular with consumers. The appearance of **the oil can be lightened by bleaching**.

h. De-gumming

De-gumming is a way of treating seed that have **high phosphatide** content.

- The phosphatide, which makes a gummy residue, **is removed by mixing the oil with 2 to 3% water.**
- This hydrated phosphatide can then be **removed by settling, filtering or centrifuged**.

PROCESSING OF ANIMAL PRODUCTS – ANIMAL, POULTRY AND FISHERIES

1. Meat And Meat Processing

Meat is **animal flesh that is used as food**. Most often, this means the skeletal muscle and associated fat and other tissues, such as organs or meat can **also be defined as "the muscle tissue of slaughter animals".**

- **Meat is composed of water, fat, protein, minerals and a small proportion of carbohydrate**.
- The lean meat **contains 20-22 percent proteins**
- The **fat content of meat varies from 5 – 40 per cent** with the type, breed, feed and age of the animal.
- Carbohydrates are found only in very small quantities in meat

Processing Of Meat

a. Curing-The curing of meat brings about the modification of meat **that affects preservation, flavor, color and tenderness due to added curing agents.**

- The prime purpose of curing is to **produce the unique flavored meat** products and a special purpose is to **preserve the red color of meat**.
- The ingredients used for curing are **common salt, sodium nitrate or nitrite, sugar and spices**.
- Nitrite **fixes the red color of myoglobin,** also a beneficial effect on the flavor of cured meats and an inhibitory effect on Clostridium botulinum.
- **Curing has also some detrimental effects during storage**. The **pink color of nitrite cured meat changes to brown, in the presence of oxygen**. Thus, cured meat should

be preferentially packed in containers from which oxygen has been excluded. **The salts of cured meat enhance oxidation of lipid components and thus reduce shelf life.**

b. Smoking-

A cured meat **may be dried and smoked**.

- Smoking also was originally used as a method of **preservation but today smoking is used mostly for its flavor contribution and coagulation of proteins**.
- Ham is frequently processed by smoking.
- The smoke contains compounds having antiseptic properties which destroy the microorganism present in meat.
- Smoking also prevents of rancidity in meat.

c. Cooking of Meat:

Cooking of meat is **done in order to improve the texture of meat**. When meat is cooked, **three types of changes contribute to increase tenderness**;

- The melting of fat,
- Dissolution of collagen in hot liquids to become soft gelatin
- Tissue softening and muscle fibre separation.

There are two methods of cooking meat- the dry heat and moist heat methods.

Dry Heat method: **Tender cuts of beef, lamb and pork** may be cooked by these methods.

1. **Roasting**: Roasting is one of the simplest and most commonly used methods of cooking meat. The meat is **adequately cooked till browning of meat for good flavor and appearance**. For small roasts, an oven **temperature of 177^{0}C is used.**

2. **Broiling:** Broiling is cooking of **meat by direct radiant heat,** such a**s the open fire of a gas flame, live coals or electric oven**. In open fire or coal broiling heat comes from below, whereas in oven broiling, the heat comes from above. The **Broiling is carried out at a temperature of 176 ^{0}C until the topside is brown.**

3. **Pan broiling**: In pan broiling, **heat is transferred to meat primarily** by conduction from **the pan or griddle**. Meat is placed in a cold griddle and heated so

that meat cooks slowly. **Pan broiling is the preferred method of cooking for thin cuts** of meats.

4. **Frying**: The **cooking of tender cuts of thin (about 1-1.5 cm thick) meat by pan frying or deep fat frying**. In pan frying, a small **amount of fat/oil is added** to the frying pan so that the melted fat is about 0.5 cm deep. In deep fat frying, the melted fat will be deep enough to cover the meat.

Moist heat Methods - Moist heat method is **used for less tender cuts of meat**.

1. **Braising:** After broiling the meat**, small amount of water is added to the browned meat and the pan used for cooking is covered** with a tight-fitting lid and cooked with a low heat the meat becomes tender.
2. **Stewing:** Large pieces of **tough cuts are cooked in water until tender**. The meat is placed in a kettle or **vessel with sufficient quantity of water** to cover the meat. The vessel is then covered and allows the water to simmers.
3. **Pressure cooking:** In pressure cooking, cooking is **done in steam at a temperature** higher than that of boiling water. Quality wise pressure-cooked meat is less juicy and cooking losses are great.

2. Poultry Meat and Egg Processing

Transportation of birds: During **loading, unloading and transport, care should be taken that birds should not injured and stressed.** Over loading must be avoided.. The crates must be easily cleanable, disinfest and durable. The birds **must be transported in the morning hours.**

Poultry Meat Processing

Hygienic slaughter of birds

Birds must be **fasted for 12 hours prior to slaughter**, but **provide albidum water,** which facilitates to **minimize the microbial load in the intestine of the birds** to reduce the risk of contamination during evisceration.

Stunning: Hang the birds for slaughter in such a way for easy stunning. By stunning the birds, maximum blood can be removed from the carcass which helps in extending the shelf life of the meat.

Bleeding: Immediately after stunning, **the birds must be stuck with a sharp knife by serving the major blood vessel sin the neck. Allow 2-3 minutes for bleeding**. Provide bleeding cones for hygienic and aesthetic removal of blood

De-feathering: **De-skinning is one of the methods and more safe method of removing the feather as the chances of microbial contamination is less in this method**. But commonly the slaughtered birds **are scaled by dipping in hot water of 50-60°C which loosens the feather and by hand picking the feathers will be removed**

Evisceration: Eviscerate the **carcass as quick as possible by removing the anal gland followed by opening the abdomen next to the keel bone** and remove the visceral organs and collect the edible and inedible offal separately for easily processing and delivery.

Washing of carcass: After evisceration, the carcass **is looked for any change in color,** or / and abnormality and the carcass without any deviation is allowed for further processing. **Remove the neck. Wash the carcass thoroughly** with potable water.

Storage: In most of the days the meat will be sold immediately**. If it doesn't happen chilling the carcass to 4°C in a chiller/refrigerator.** In case of small processors keep the meat in the insulated boxes with ice.

3. Egg Processing

Egg has a **shell made of calcium carbonate** which **is white or brown in colour**.

Eggs contain two parts albumen **white part (56-61 per cent) and yellow part yolk (27.30 per cent)**.

Egg contains **12-14 per cent proteins**, with all the essential amino acids.

Processing Of Egg

Egg drying: Egg powder is **prepared by drying the egg**. Egg powder may be **reconstituted before use by mixing in desired quantity of water.** The white, yolk or whole egg after pasteurization may be dried with the help of different methods of drying. **Spray drying method is commonly used for drying the eggs**.

The **powdered egg is cooled and packed** in suitable packaging material.

Egg freezing: Egg may be **preserved by freezing**. In freezing **egg cannot be frozen in the shell since the egg shell would crack with the expansion of liquids when frozen**.

Therefore, the **whole of the egg liquid content is frozen or it is separated into white and yolk and then frozen**.

The **whole egg or yolk is pasteurized for 3.5 to 4 min**. without changes in physical and functional properties.

Pasteurized whole or **separated eggs are placed in suitable container and** frozen in sharp freezer room with **circulating air at -29 ^{0}C.** The freezing process may be completed in 48-72 hours.

4. Fish Processing

About 30 per cent fish processed for human consumption is marketed as fresh; **the remainder is frozen fish and fillets in ready-to-eat meals and other convenience products**. As the fish and **sea food is perishable**, it is very essential to process and thereby it can be stored for longer period.

Primary Processing of Fish and Sea Food

Pretreatment: Fish are kept on ice in boxes before delivery to the processing plant. **Pretreatment involves ice removal, washing, grading according to size and de-heading.** Some fish are **skinned by immersing in to a warm caustic bath**.

Filleting: Filleting is performed by machines with **mechanical knives that cut the fillets from the backbone and remove the collarbone.** Some fillets may be skinned at this step in this process.

Trimming and Inspection: In the trimming section, **pin bones are removed and operators inspect the fillets**. Any **defects and any inferior parts found are removed**. Off cuts are collected and minced. Depending upon the final products, the fillets can be cut in to portions according to weight or final product requirements.

Storage/Packaging (Fresh): Fresh products **are packaged in boxes with ice which is separated from the product by a layer of plastic.**

Preservation Methods of Fish

Fish is a perishable commodity. It cannot be store in **normal condition for longer period**.

There are some in practice to preserve fish which is explained here under.

Canning of Fish

Canning is performed by two methods: pre-cooking and raw pack.

Precooking begins with thawing of the fish which are then eviscerated, washed and cooked. Canning retains the natural flavor of the fish usually oily fish are most suitable for canning. Cooking occurs **with steam, oil, hot air or smoke for up to 10** hours, depending upon the fish size.

After cooling, the head, fins, bones, and undesirable meats are removed and remainder is chopped / cut and placed in cans.

Additional fish or **vegetable oil, brine, and/or water are added to the cans which are sealed and pressure-cooked before shipment**.

Finally, the cans **are sealed and sterilized with hot water or steam** and then stored.

Smoking Of Fish

Smoking is a process of **treating fish by exposing it to smoke** from smouldering wood or plant materials to **introduce flavour, taste, and preservative ingredients into the fish**. This process is usually characterised by an integrated combination of salting, drying, heating and smoking steps in a smoking chamber. For smoked fish and fishery products, a minimum **thermal process of 30 min at or above 145 °F (62.8 °C) is required** by FDA (2001).

Pickling (Salting) of Fish

The dry salting or wet salting method of pickling is widely used for fish processing in India.

In dry salting method, first fishes are rubbed with salt powder and then packed in tubs with dry **salt powder sprinkled** in between layers of fishes. After 15-20 hours, the fishes are removed from tubs, washed in **brine and then followed by sun drying for 2-3 days**.

In wet salting cleaned fishes with longitudinal silts (cut) are packed in large container which containing concentrated salt solutions and stirred daily till properly pickled. Wet salted (pickled) fish is sold into market without drying

Freezing of Fish

Freezing, greatly extend the period of storage. In freezing, if the fish is **frozen down to -28 to -30^{0}C within two hours** of its catch gives effective keeping quality as similar to that of fresh fish

Drying of Fish

Sun drying is the most widely used method for drying. Drying removes moisture from tissues and arrests the bacterial and enzymatic growth.

Fish Meal and Fish Oil Production

Fish meal - is delivered from the dry components of the fish and the oil from oily component. The **meal is used in animal and pet feed** due to its high protein content.

Fish Oil- **is separated from the liquid by centrifuge** and **is polished by using hot water washes** and additional centrifuging. The **removed water is evaporated to concentrate the solids and the remaining oil is refined to remove any impurities**.

Fish Flour (Fish Protein Concentrate)

Fish flour **contains 85 to 90 per cent of high-quality protein** The fat extracted **tissue is dried after evaporation of solvent by dehydration process using dryer.**

After **dehydration, milling gives a bland highly nutritious powder rich in high quality protein and minerals**.